Factors Considered When Communicating a Possible Cancer Warning and How They Can Be Used in the App

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1. Introduction

When a doctor communicates a serious warning such as a possible cancer-related finding, the message is not based only on the medical result. Doctors also consider the emotional, social, economic, and psychological condition of the patient. These same ideas can be applied to the design of a mobile health application such as SkinSafe.

This document explains the main factors involved in doctor-patient communication and shows how each factor can be translated into practical app design decisions.

2. Mental State of the Patient

Why it matters

If a patient is already anxious, frightened, or emotionally overwhelmed, harsh or direct wording can increase panic. For this reason, doctors usually communicate carefully, observe the patient's reaction, and explain serious findings gradually.

How it can be used in the app

The app should avoid shocking or panic-inducing language. Instead of using phrases such as "Cancer Detected" or "Dangerous Result," the app should use calm and action-focused wording.

Practical app design ideas

- Use supportive result headings such as "Caution Advised" or "Specialist Review Recommended."

- Avoid flashing red warning visuals or aggressive alert language.
- Present the result in stages: short summary, simple explanation, and next action.
- Use soft and medically calm colors.
- Add supportive wording such as “This result is intended to guide your next step.”

3. Economic and Socioeconomic Factors

Why it matters

Even if the doctor gives the correct warning, the patient may not be able to immediately afford specialist consultation, laboratory tests, transport, or time away from work. Therefore, doctors often think practically as well as clinically.

How it can be used in the app

The app should not only give a warning. It should also suggest realistic next steps that are manageable for different users.

Practical app design ideas

- Allow the user to save the result and monitor the same lesion over time.
- Allow repeat scanning and comparison of previous records.
- Use guidance such as “Repeat this scan in a few days if the lesion changes.”
- Avoid forcing only one costly or urgent path unless the risk is clearly high.
- Help users prepare history and images before seeing a doctor.

4. Health Literacy

Why it matters

Many patients may not understand technical medical words such as melanoma, malignant, lesion, or biopsy. Doctors often simplify language depending on the patient’s educational and medical understanding.

How it can be used in the app

The app should explain results using simple language. Technical words should be minimized or clearly explained.

Practical app design ideas

- Add a “What this means” section under the result.
- Use simple explanations under medical terms.
- Teach the ABCDE rule with easy cards and examples.
- Show simple information first, and detailed information only if the user wants more.
- Use patient-friendly terms such as “higher concern pattern” instead of overly technical labels.

5. Family Support and Social Context

Why it matters

Some patients cope better when a family member or trusted supporter is involved. Others prefer private review first. Doctors often adapt communication based on the patient’s support system and personal preference.

How it can be used in the app

The app can allow the user to save, review, and later share the result. This supports discussion with a family member or doctor without forcing an immediate emotional decision.

Practical app design ideas

- Allow users to save the result in history.
- Provide a summary that can be shown later during consultation.
- Add wording such as “You may wish to discuss this result with a family member or clinician.”
- Support future report-sharing features.
- Avoid forcing the user to react instantly on the same screen.

6. Privacy, Setting, and Timing

Why it matters

Doctors do not normally give serious news in an exposed, noisy, or rushed environment. Privacy and timing influence how safely the patient receives the information.

How it can be used in the app

The result screen should feel calm, private, and focused. Sensitive results should not be shown in a visually chaotic or alarm-heavy way.

Practical app design ideas

- Use a clean and uncluttered result layout.
- Reduce unnecessary distractions on the result page.
- Keep the tone respectful and calm.
- Use progressive disclosure instead of too much information at once.
- Present next steps clearly and calmly.

7. Patient Preference and Readiness

Why it matters

Not every patient wants the same amount of information immediately. Some prefer a short summary first. Others want detailed explanation and probabilities.

How it can be used in the app

The app should let users move from basic information to detailed information gradually.

Practical app design ideas

- Show a simple summary first.
- Add optional detailed explanation below.
- Use clear sections such as “Result Summary,” “What This Means,” and “What to Do Next.”
- Keep high-level advice visible even if technical details are hidden.

8. The SPIKES Protocol and Its Application in the App

The SPIKES protocol is a widely used framework for breaking bad news in a structured and empathetic way.

SPIKES stands for:

- **S** – Setting
- **P** – Perception
- **I** – Invitation
- **K** – Knowledge
- **E** – Empathy
- **S** – Strategy and Summary

How SPIKES can be translated into the app

- **Setting:** Use a calm and respectful result screen.
- **Perception:** Use the questionnaire to understand what the user has already noticed.
- **Invitation:** Show a short summary first, then allow the user to read more details.
- **Knowledge:** Present the risk result in simple language.
- **Empathy:** Use soft and supportive wording.
- **Strategy and Summary:** End with practical next steps such as saving history, repeating the scan, or seeking specialist review.

Using the SPIKES idea helps transform the app from a simple image-classification tool into a more ethical and patient-centered digital health system.

9. Summary Table

Factor	Why Doctors Consider It	Example App Response
Mental state	To avoid panic and emotional overload	Calm wording, soft UI, supportive result summary
Economic factors	To make next steps realistic and practical	Save history, follow-up guidance, repeat scan options
Health literacy	To ensure understanding	Plain language, simplified medical explanation
Family support	To improve coping and follow-up	Save/share-ready summary, history review
Privacy and setting	To protect dignity and reduce stress	Clean result screen, no alarming interface
Readiness and preference	To match user information needs	Short summary first, details later
SPIKES structure	To communicate difficult findings ethically	Guided result flow with empathy and action plan

10. Conclusion

The factors doctors consider when communicating a possible cancer warning can be directly applied to a mobile application. In SkinSafe, these factors are not only medical communication ideas; they become practical design principles.

By considering mental state, economic limitations, health literacy, family support, privacy, and SPIKES-based communication structure, the application can present results in a more ethical, human-centered, and clinically respectful way.

This gives the project an important research contribution beyond AI prediction alone. It shows that the system is designed not only to analyze lesions, but also to communicate risk responsibly.